

Discrete-Time Signals

May 13, 2026

What is a Discrete-Time Signal?

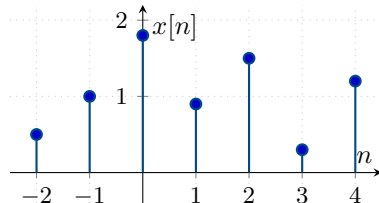
Definition

A **discrete-time signal** $x[n]$ is a function defined only at integer values of $n \in \mathbb{Z}$, i.e. a sequence of numbers:

$$x[n] = \{\dots, x[-1], x[0], x[1], x[2], \dots\}$$

- n is the **sample index**
- Signal is *undefined* between integer values of n
- Represented as a **stem plot**

Generic Discrete-Time Signal



Discrete-Time Signals

Discrete-Time Signal

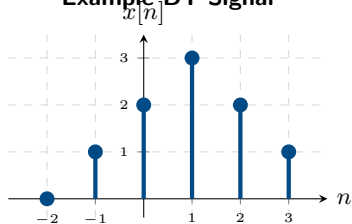
A **DT signal** $x[n]$ is a function defined only at *integer* values $n \in \mathbb{Z}$. Between integers the signal is **undefined**.

Three equivalent representations:

- 1 **Functional** — closed-form mathematical expression
- 2 **Tabular** — table listing n vs. $x[n]$
- 3 **Sequence** — ordered list with origin marker $\uparrow$

$$x : \mathbb{Z} \rightarrow \mathbb{R}, \quad x[n] \text{ defined } \forall n \in \mathbb{Z}$$

Example DT Signal



Samples exist only at integer n ; the dots are the signal.

1 — Functional Representation

Standard building blocks:

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases} \quad u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

Derived signals:

- Real exponential: $x[n] = A a^n u[n]$
- Sinusoidal: $x[n] = A \cos(\omega_0 n + \phi)$
- Rectangular: $x[n] = u[n] - u[n - N]$

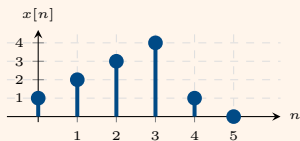
Key Advantage

One formula describes *infinitely many* samples compactly.

Functional Representation — Further Examples

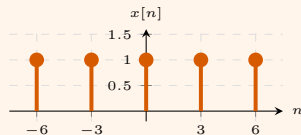
Piecewise Signal

$$x[n] = \begin{cases} n + 1, & 0 \leq n \leq 3 \\ 5 - n, & 4 \leq n \leq 5 \\ 0, & \text{otherwise} \end{cases}$$



Periodic Impulse Train

$$x[n] = \sum_{k=-\infty}^{\infty} \delta[n - 3k], \quad N = 3$$



2 — Tabular Representation

Definition

List sample values in a **two-row table**: one row for the index n , one row for the amplitude $x[n]$.

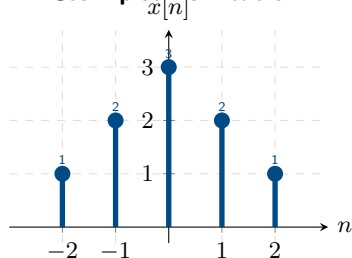
When to use:

- Finite-length or short signals
- Verifying computed values
- Simulation / numerical work

Example: triangular pulse

n	-2	-1	0	1	2
$x[n]$	1	2	3	2	1

Stem plot from table



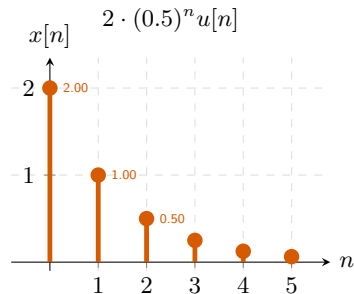
Tabular Representation — Example

Signal: $x[n] = 2 \cdot (0.5)^n u[n]$

Table of values

n	0	1	2	3	4	5	...
$x[n]$	2	1	0.50	0.25	0.13	0.06	...

- Table shows a **finite window** of an infinite signal



3 — Sequence Representation

Definition

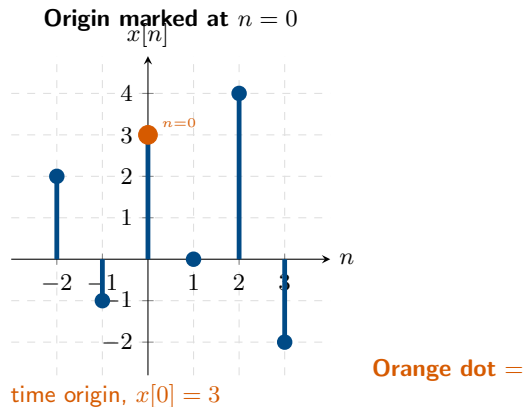
Write $x[n]$ as an ordered list of values. The **time origin** ($n = 0$) is marked with an upward arrow $\uparrow$ placed below the corresponding sample.

General notation:

$$x[n] = \{\dots, x_{-2}, x_{-1}, x_0, x_1, x_2, \dots\}$$

Example

$$x[n] = \{2, -1, 3, 0, 4, -2\}$$



Sequence Representation — Origin Position Matters

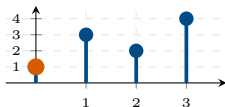
Key Rule

The *same values* yield **different signals** depending on where the $\uparrow$ is placed. Always state the origin explicitly.

Case A — origin at 1st sample:

$$x_A[n] = \{1, 3, 2, 4\}$$

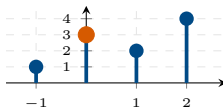
$\uparrow$



Case B — origin at 2nd sample:

$$x_B[n] = \{1, 3, 2, 4\}$$

$\uparrow$



Relationship

Shifting the origin = **time shift** of the signal.

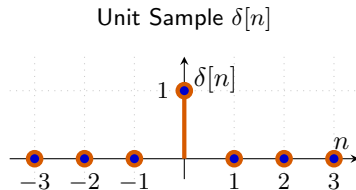
$$x_B[n] = x_A[n + 1]$$

Some Elementary Discrete-Time Signals

Unit Sample Sequence $\delta[n]$

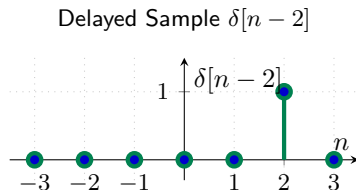
Definition

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$



Delayed Impulse

$$\delta[n - k] = \begin{cases} 1, & n = k \\ 0, & n \neq k \end{cases}$$



- Nonzero only at one sample
- Used to represent discrete-time signals
- Also called the **discrete impulse**

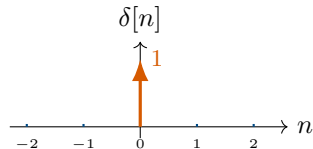
What is the Unit Impulse (Delta) Signal?

Definition: Unit Impulse $\delta[n]$

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & \text{otherwise} \end{cases}$$

Key Properties:

- Single non-zero value at $n = 0$
- Also called a “spike” or Dirac-like sequence
- The 0th sample equals 1



Shifted impulse: $\delta[n - k]$ is 1 only at $n = k$, zero elsewhere.

Scaled & Shifted Impulse

Notation: $m \delta[n + k]$

- m — **scale** (amplitude multiplier)
- k — **shift** (time index where spike occurs)

$$m \delta[n + k] = m \quad \text{at } n = -k, \quad 0 \text{ otherwise}$$

Examples:

$$3 \delta[n - 2] \Rightarrow \text{spike of height 3 at } n = 2$$

$$-2 \delta[n + 1] \Rightarrow \text{spike of height } -2 \text{ at } n = -1$$

Time Shifting

Delay (Right Shift)

$$y[n] = x[n - k], \quad k > 0$$

Signal moves **right** by k samples.
(Introduces time delay)

Advance (Left Shift)

$$y[n] = x[n + k], \quad k > 0$$

Signal moves **left** by k samples.
(Advances in time)

Quick Check

If $x[n] = \{1, 2, 3\}$ starting at $n = 0$, then
 $x[n - 2] = \{1, 2, 3\}$ starting at $n = 2$.

Every Signal as a Sum of Impulses

Core Theorem

*“Every signal is a summation of a bunch of **shifted & scaled** Unit Impulses.”*

Formal expression:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k]$$

Each term $x[k] \delta[n - k]$ is a scaled, shifted impulse that “picks out” the value $x[k]$ at time k .

Unit Sample – Worked Example

Example 1

Express the signal $x[n] = \{2, 0, -1, 3\}$ starting at $n = -1$ as a sum of scaled impulses.

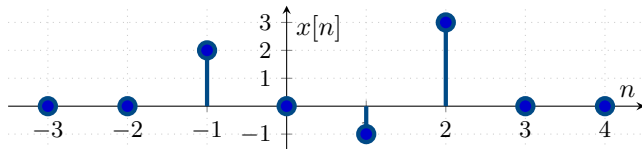
Unit Sample – Worked Example

Solution

The non-zero samples are at $n = -1, 0, 1, 2$ with values 2, 0, -1, 3.

$$x[n] = 2\delta[n+1] + 0 \cdot \delta[n] + (-1)\delta[n-1] + 3\delta[n-2]$$

$$x[n] = 2\delta[n+1] - \delta[n-1] + 3\delta[n-2]$$



Step-by-Step Decomposition Example

Given signal with origin at $n = 0$:

$$x[n] = \{2, \underset{\uparrow}{4}, 0, 3\}$$

Step-by-Step Decomposition Example

Given signal with origin at $n = 0$:

$$x[n] = \{2, \underset{\uparrow}{4}, 0, 3\}$$

Step 1: Extract each sample as a scaled, shifted impulse:

$$\{2, 0, 0, 0\} = 2\delta[n + 1]$$

$$\{0, 4, 0, 0\} = 4\delta[n]$$

$$\{0, 0, 0, 0\} = 0\delta[n - 1]$$

$$\{0, 0, 0, 3\} = 3\delta[n - 2]$$

Step-by-Step Decomposition Example

Given signal with origin at $n = 0$:

$$x[n] = \{2, \underset{\uparrow}{4}, 0, 3\}$$

Step 1: Extract each sample as a scaled, shifted impulse:

$$\{2, 0, 0, 0\} = 2\delta[n + 1]$$

$$\{0, 4, 0, 0\} = 4\delta[n]$$

$$\{0, 0, 0, 0\} = 0\delta[n - 1]$$

$$\{0, 0, 0, 3\} = 3\delta[n - 2]$$

Step 2: Sum them up:

$$x[n] = 2\delta[n + 1] + 4\delta[n] + 0\delta[n - 1] + 3\delta[n - 2]$$

$\Rightarrow$ Verified: $\{2, 4, 0, 3\}$ ✓

Definition

Impulse decomposition breaks an N samples signal into N component signals, each containing N samples. Each of the component signals contains one point from the original signal, with the remainder of the values being zero. A single nonzero point in a string of zeros is called an impulse.

Impulse Decomposition

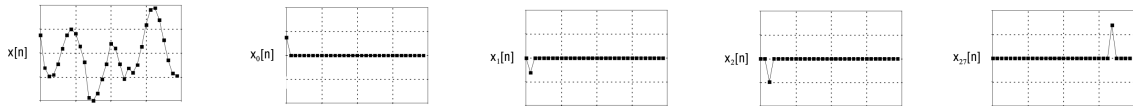


Figure: Impulse Decomposition

Unit Step Sequence $u[n]$

Definition

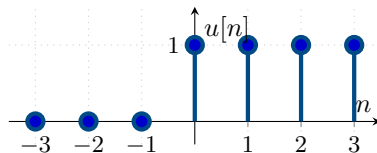
$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

Relation to impulse:

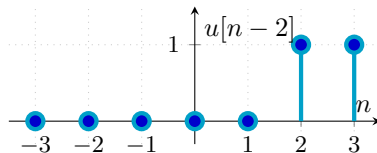
$$u[n] = \sum_{k=0}^{\infty} \delta[n-k] = \sum_{k=-\infty}^n \delta[k]$$

$$\delta[n] = u[n] - u[n-1]$$

Unit Step $u[n]$



Delayed Step $u[n-2]$



Unit Step – Worked Example

Example 2

Express the rectangular pulse $x[n] = 1$ for $2 \leq n \leq 5$, zero elsewhere, using unit step functions.

Unit Step – Worked Example

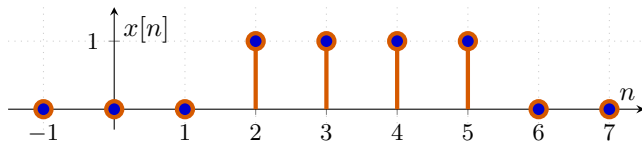
Solution

A pulse that *starts* at $n = 2$ is $u[n - 2]$.

A pulse that *ends* at $n = 5$ means subtracting $u[n - 6]$:

$$x[n] = u[n - 2] - u[n - 6]$$

Verify: for $n = 2, 3, 4, 5$: $1 - 0 = 1$. For $n \geq 6$: $1 - 1 = 0$. ✓



Unit Ramp Sequence $r[n]$

Definition

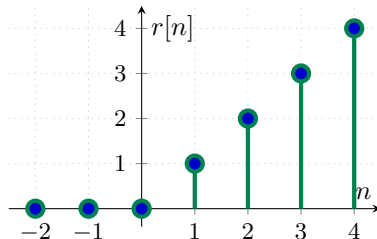
$$r[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

Relations:

$$r[n] = \sum_{k=0}^{\infty} u[n-k] = \sum_{m=-\infty}^n u[m]$$

$$u[n] = r[n] - r[n-1]$$

Unit Ramp $r[n] = n u[n]$



Exponential Sequence $x[n] = A a^n$

Definition

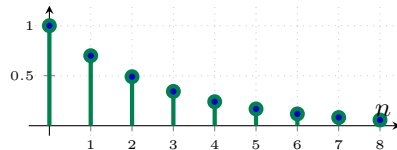
$$x[n] = A a^n, \quad n \in \mathbb{Z}$$

where $A \in \mathbb{R}$ (amplitude) and $a \in \mathbb{R}$ (base).

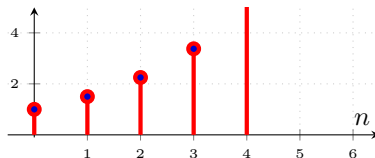
Behaviour by $|a|$:

- $|a| < 1$: **decaying** envelope
- $|a| > 1$: **growing** envelope
- $|a| = 1$: **constant** magnitude ($a = \pm 1$)
- $a < 0$: signal **alternates** in sign

Decaying: $a = 0.7$



Growing: $a = 1.5$



Symmetric (Even) Signals

Definition – Even Signal

A signal $x[n]$ is **even** (symmetric) if

$$x[-n] = x[n] \quad \forall n \in \mathbb{Z}$$

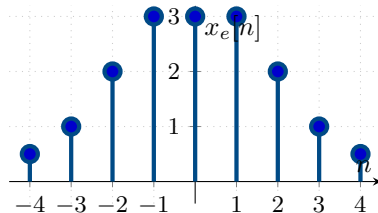
Even part of any signal:

$$x_e[n] = \frac{x[n] + x[-n]}{2}$$

Properties:

- Mirror symmetry about $n = 0$
- $\sum_{n=-\infty}^{\infty} x_e[n]$ is always real (if x is real)
- $\cos(\omega_0 n)$ is even

Even Signal Example



Anti-Symmetric (Odd) Signals

Definition – Odd Signal

A signal $x[n]$ is **odd** (anti-symmetric) if

$$x[-n] = -x[n] \quad \forall n \in \mathbb{Z}$$

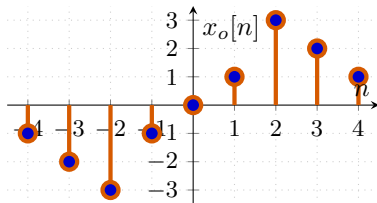
Odd part of any signal:

$$x_o[n] = \frac{x[n] - x[-n]}{2}$$

Properties:

- **Always** $x_o[0] = 0$
- $\sin(\omega_0 n)$ is odd
- $\sum_{n=-\infty}^{\infty} x_o[n] = 0$

Odd Signal Example



Even–Odd Decomposition Theorem

Theorem

Every real signal $x[n]$ can be *uniquely* decomposed as:

$$x[n] = x_e[n] + x_o[n]$$

where

$$x_e[n] = \frac{x[n] + x[-n]}{2} \quad (\text{even part}), \quad x_o[n] = \frac{x[n] - x[-n]}{2} \quad (\text{odd part})$$

Quick checks:

- $x_e[-n] = +x_e[n]$ ✓ (even symmetry preserved)
- $x_o[-n] = -x_o[n]$ ✓ (odd symmetry preserved)
- $x_e[n] + x_o[n] = \frac{x[n] + x[-n]}{2} + \frac{x[n] - x[-n]}{2} = x[n]$ ✓

Folding (Reflection)

Time Reversal / Folding

$$y[n] = x[-n]$$

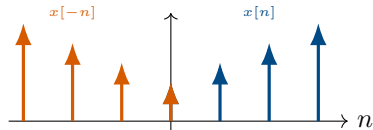
Reflects the signal about the **vertical axis** ($n = 0$).

Also called *folding* — a **virtual** operation (not physical delay).

Example:

$x[n] = \{1, 2, 3, 4\}$ at $n = 0, 1, 2, 3$

$x[-n] = \{4, 3, 2, 1\}$ at $n = -3, -2, -1, 0$



Combined Transformation: $x[-n + 2]$

Problem: Find $x(-n + 2)$ from $x[n]$

Two valid interpretations — **Order Matters!**

Method A (standard):

① Reflect: $x[n] \rightarrow x[-n]$

② Shift left by 2:

$$x[-n] \rightarrow x[-(n - 2)] = x[-n + 2]$$

Net effect: **right shift by 2**

Clarification:

$$x[-(n - 2)] = x[-n + 2]$$

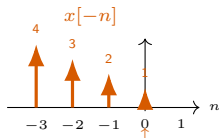
Note: $x[-(n - 2)]$ means replace n with $n - 2$ *after* folding, which shifts right.

Combined Transformation: $x[-n + 2]$

Problem: given $x[n]$

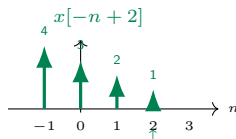
Two approaches — **Order Matters!**

Step 1: Reflect $x[n] \rightarrow x[-n]$

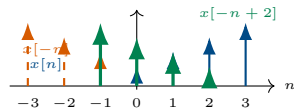


Step 2: Shift right by 2

$$x[-n] \rightarrow x[-(n - 2)] = x[-n + 2]$$



All three overlaid



Down Sampling

Definition: Down Sampling by factor M

$$y[n] = x[Mn]$$

Keep every M^{th} sample of $x[n]$.

For $M = 2$: $y[n] = x[2n]$

n (original)	...	0	1	2	3	4
$x[n]$	...	$x[0]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$
n (down-sampled)	...	0		1		2
$y[n] = x[2n]$	...	$x[0]$		$x[2]$		$x[4]$

The 2nd sample of original $\rightarrow$ 1st sample of down-sampled signal.
Operates on a *sample-to-sample* basis.

Total Energy of a Discrete-Time Signal

$$E_N = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

The $|\cdot|^2$ handles **complex-valued** signals (takes magnitude squared).

Real signals:

$$E_N = \sum_n x[n]^2$$

Complex signals ($x[n] = a + jb$):

$$|x[n]|^2 = a^2 + b^2$$

Example

$$x[n] = \{2, 4, 0, 3\}$$

$$E_N = 2^2 + 4^2 + 0^2 + 3^2 = 4 + 16 + 0 + 9 = \mathbf{29}$$

Periodic vs. Aperiodic Signals

A discrete-time signal $x[n]$ is **periodic** with period N if:

$$x[n + N] = x[n] \quad \forall n, \quad N \in \mathbb{Z}^+$$

Frequency Condition for Periodicity

For $x[n] = A \cos(2\pi f_0 n + \phi)$, the signal is periodic iff

$$f_0 = \frac{K}{N}, \quad K, N \in \mathbb{Z}$$

i.e., f_0 is a **rational number**.

Rational $f_0 \Rightarrow$ Periodic

Example: $f_0 = \frac{1}{4}$, period $N = 4$

Irrational $f_0 \Rightarrow$ Aperiodic

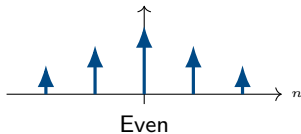
Examples: $f_0 = 3.33\bar{2}$, $1.71\bar{4}$

Even and Odd Signals

Even Signal

$$x[n] = x[-n]$$

Symmetric about $n = 0$.

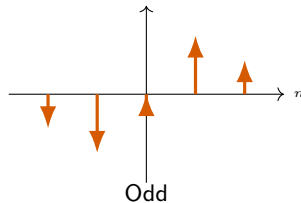


Odd Signal

$$x[n] = -x[-n]$$

Anti-symmetric about $n = 0$.

Note: $x[0] = 0$ always.



Even–Odd Decomposition

Theorem

*“Any discrete-time signal can be expressed as the summation of **one even** and **one odd** component.”*

$$x[n] = x_e[n] + x_o[n]$$

Even–Odd Decomposition

Theorem

*“Any discrete-time signal can be expressed as the summation of **one even** and **one odd** component.”*

$$x[n] = x_e[n] + x_o[n]$$

Derivation:

$$x[n] = x_e[n] + x_o[n] \tag{1}$$

$$x[-n] = x_e[n] - x_o[n] \tag{2 (substitute $-n$)}$$

Even–Odd Decomposition

Theorem

*“Any discrete-time signal can be expressed as the summation of **one even** and **one odd** component.”*

$$x[n] = x_e[n] + x_o[n]$$

Derivation:

$$x[n] = x_e[n] + x_o[n] \tag{1}$$

$$x[-n] = x_e[n] - x_o[n] \tag{2 (substitute $-n$)}$$

Adding and subtracting (1) and (2):

$$x_e[n] = \frac{1}{2} [x[n] + x[-n]] \quad x_o[n] = \frac{1}{2} [x[n] - x[-n]]$$

Even–Odd Decomposition: Problem Statement

Example

Decompose the discrete-time signal

$$x[n] = \{1, 2, 3, 4\}$$

where

$$n = 0, 1, 2, 3$$

into:

$$x_e[n] \quad \text{and} \quad x_o[n]$$

Step 1: Find the Folded Signal $x[-n]$

Original signal:

$$x[n] = \{1, 2, 3, 4\}$$

with indices:

$$n = 0, 1, 2, 3$$

Time Reversal (Folding)

Flip the signal around $n = 0$:

$$x[-n] = \{4, 3, 2, 1\}$$

with indices:

$$n = -3, -2, -1, 0$$

Step 2: Compute the Even Part

Formula:

$$x_e[n] = \frac{1}{2} (x[n] + x[-n])$$

Example Calculations

$$x_e[0] = \frac{x[0] + x[0]}{2} = \frac{1 + 1}{2} = 1$$

$$x_e[1] = \frac{x[1] + x[-1]}{2} = \frac{2 + 0}{2} = 1$$

$$x_e[-1] = \frac{x[-1] + x[1]}{2} = \frac{0 + 2}{2} = 1$$

Step 3: Compute the Odd Part

Formula:

$$x_o[n] = \frac{1}{2} (x[n] - x[-n])$$

Example Calculations

$$x_o[0] = \frac{1 - 1}{2} = 0$$

$$x_o[1] = \frac{2 - 0}{2} = 1$$

$$x_o[-1] = \frac{0 - 2}{2} = -1$$

Observation:

$$x_o[n] = -x_o[-n]$$

So, the odd part is antisymmetric.